Advanced Topics in ACP

https://taroccoesbrocco.github.io/ACP2026.html

Solutions to selected exercises from Lecture 2 on the λ-calculus

Giulio Guerrieri

Institut für Informatik, Universität Innsbruck. giulio.guerrieri@uibk.ac.at

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Non-assessed exercises from Lecture 2

(https://taroccoesbrocco.github.io/ACP2026/Lecture02.pdf)

Exercise 1

Find the simply typed λ-terms (in Curry-style and Church-style) associated with the derivations in ND found for the facts below (see Exercise 1 from Lecture 1).

1. $\vdash X \Rightarrow ((X \Rightarrow Y) \Rightarrow Y)$.
2. $((X \Rightarrow Y) \Rightarrow X \Rightarrow Z)^z \vdash Y \Rightarrow X \Rightarrow Z$.
3. $((X \Rightarrow Y) \Rightarrow X)^z \vdash Y \Rightarrow X$.
4. $(X \Rightarrow Y \Rightarrow Z)^z \vdash Y \Rightarrow X \Rightarrow Z$.
5. $(X \Rightarrow Y \Rightarrow Z)^z, (X \Rightarrow Y)^y \vdash X \Rightarrow Z$.
6. $((X \Rightarrow X) \Rightarrow Y)^y \vdash (Y \Rightarrow Z) \Rightarrow Z$.

Solution to Exercise 1

1. In Curry-style and Church-style for λ-terms, and ND for derivations:

$$\dfrac{\dfrac{\dfrac{[y : (X \Rightarrow Y)^y] \quad [x : X^x]}{yx : Y}\Rightarrow_e}{\lambda y.yx : (X \Rightarrow Y) \Rightarrow Y}\Rightarrow_i^y}{\lambda x.\lambda y.yx : X \Rightarrow ((X \Rightarrow Y) \Rightarrow Y)}\Rightarrow_i^x \qquad \dfrac{\dfrac{\dfrac{[y : (X \Rightarrow Y)^y] \quad [x : X^x]}{yx : Y}\Rightarrow_e}{\lambda y^{X\Rightarrow Y}.yx : (X \Rightarrow Y) \Rightarrow Y}\Rightarrow_i^y}{\lambda x^X.\lambda y^{X\Rightarrow Y}.yx : X \Rightarrow ((X \Rightarrow Y) \Rightarrow Y)}\Rightarrow_i^x$$

2. In Curry-style and Church-style for λ-terms, and ND for derivations:

$$\dfrac{\dfrac{z : ((X \Rightarrow Y) \Rightarrow X \Rightarrow Z)^z \quad \dfrac{[y : Y^y]}{\lambda x.y : X \Rightarrow Y}\Rightarrow_i^x}{z(\lambda x.y) : X \Rightarrow Z}\Rightarrow_e}{\lambda y.z(\lambda x.y) : Y \Rightarrow X \Rightarrow Z}\Rightarrow_i^y \qquad \dfrac{\dfrac{z : ((X \Rightarrow Y) \Rightarrow X \Rightarrow Z)^z \quad \dfrac{[y : Y^y]}{\lambda x^X.y : X \Rightarrow Y}\Rightarrow_i^x}{z(\lambda x^X.y) : X \Rightarrow Z}\Rightarrow_e}{\lambda y^Y.z(\lambda x^X.y) : Y \Rightarrow X \Rightarrow Z}\Rightarrow_i^y$$

3. In Curry-style and Church-style for λ-terms, and ND for derivations:

$$\dfrac{\dfrac{z : ((X \Rightarrow Y) \Rightarrow X)^z \quad \dfrac{[y : Y^y]}{\lambda x.y : X \Rightarrow Y}\Rightarrow_i^x}{z(\lambda x.y) : X}\Rightarrow_e}{\lambda y.z(\lambda x.y) : Y \Rightarrow X}\Rightarrow_i^y \qquad \dfrac{\dfrac{z : ((X \Rightarrow Y) \Rightarrow X)^z \quad \dfrac{[y : Y^y]}{\lambda x^X.y : X \Rightarrow Y}\Rightarrow_i^x}{z(\lambda x^X.y) : X}\Rightarrow_e}{\lambda y^Y.z(\lambda x^X.y) : Y \Rightarrow X}\Rightarrow_i^y$$

4. In Curry-style and Church-style for λ-terms, and ND for derivations:

$$\dfrac{\dfrac{\dfrac{z : (X \Rightarrow Y \Rightarrow Z)^{z} \qquad [x : X^{x}]}{zx : Y \Rightarrow Z} \Rightarrow_{e} \qquad [y : Y^{y}]}{zxy : Z} \Rightarrow_{e}}{\dfrac{\lambda x.zxy : X \Rightarrow Z}{\lambda y.\lambda x.zxy : Y \Rightarrow X \Rightarrow Z} \Rightarrow_{i}^{y}} \Rightarrow_{i}^{x} \qquad \dfrac{\dfrac{\dfrac{z : (X \Rightarrow Y \Rightarrow Z)^{z} \qquad [x : X^{x}]}{zx : Y \Rightarrow Z} \Rightarrow_{e} \qquad [y : Y^{y}]}{zxy : Z} \Rightarrow_{e}}{\dfrac{\lambda x^{X}.zxy : X \Rightarrow Z}{\lambda y^{Y}.\lambda x^{X}.zxy : Y \Rightarrow X \Rightarrow Z} \Rightarrow_{i}^{y}} \Rightarrow_{i}^{x}$$

5. In Curry-style and Church-style for λ-terms, and ND for derivations:

$$\dfrac{\dfrac{\dfrac{z : (X \Rightarrow Y \Rightarrow Z)^{z} \quad [x : X^{x}]}{zx : Y \Rightarrow Z} \Rightarrow_{e} \quad \dfrac{y : (X \Rightarrow Y)^{y} \quad [x : X^{x}]}{yx : Y} \Rightarrow_{e}}{zx(yx) : Z} \Rightarrow_{e}}{\lambda x.zx(yx) : X \Rightarrow Z} \Rightarrow_{i}^{x} \qquad \dfrac{\dfrac{\dfrac{z : (X \Rightarrow Y \Rightarrow Z)^{z} \quad [x : X^{x}]}{zx : Y \Rightarrow Z} \Rightarrow_{e} \quad \dfrac{y : (X \Rightarrow Y)^{y} \quad [x : X^{x}]}{yx : Y} \Rightarrow_{e}}{zx(yx) : Z} \Rightarrow_{e}}{\lambda x^{X}.zx(yx) : X \Rightarrow Z} \Rightarrow_{i}^{x}$$

6. In Curry-style and Church-style for λ-terms, and ND for derivations:

$$\dfrac{\dfrac{[z : (Y \Rightarrow Z)^{z}] \qquad \dfrac{y : ((X \Rightarrow X) \Rightarrow Y)^{y} \quad \dfrac{[x : X^{x}]}{\lambda x.x : X \Rightarrow X} \Rightarrow_{i}^{x}}{y\,\lambda x.x : Y} \Rightarrow_{e}}{z(y\,\lambda x.x) : Z} \Rightarrow_{e}}{\lambda z.z(y\,\lambda x.x) : (Y \Rightarrow Z) \Rightarrow Z} \Rightarrow_{i}^{z} \qquad \dfrac{\dfrac{[z : (Y \Rightarrow Z)^{z}] \qquad \dfrac{y : ((X \Rightarrow X) \Rightarrow Y)^{y} \quad \dfrac{[x : X^{x}]}{\lambda x^{X}.x : X \Rightarrow X} \Rightarrow_{i}^{x}}{y\,\lambda x^{X}.x : Y} \Rightarrow_{e}}{z(y\,\lambda x^{X}.x) : Z} \Rightarrow_{e}}{\lambda z^{Y \Rightarrow Z}.z(y\,\lambda x^{X}.x) : (Y \Rightarrow Z) \Rightarrow Z} \Rightarrow_{i}^{z}$$

Exercise 2

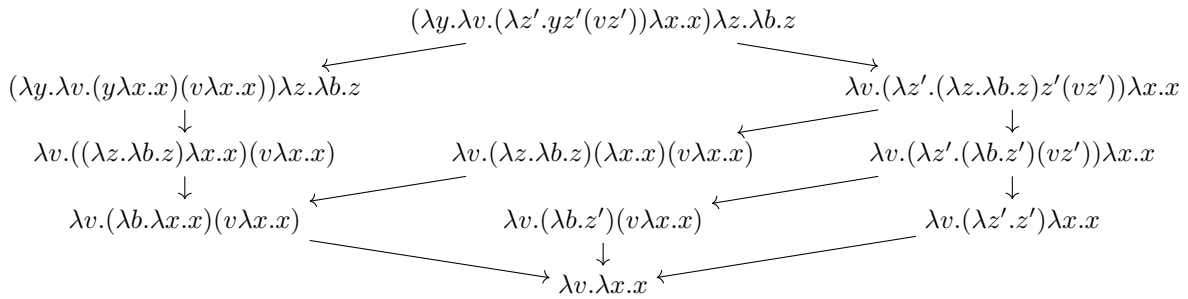
Perform all possible β-reduction steps from the λ-term decorating the derivation $\mathcal{D}$ in ND on p. 31 of Lecture 1, until you get a β-normal form. Is it always the same? Compare it with the normal derivation obtained by cut-elimination steps from $\mathcal{D}$.

Solution to Exercise 2

The derivation on p. 31 of Lecture 1 slides is $\mathcal{D}$ below, decorated with λ-terms is Curry-style.

$$\dfrac{\dfrac{\dfrac{\dfrac{[y : ((X \Rightarrow X) \Rightarrow B \Rightarrow X \Rightarrow X)^{y}] \quad [z' : (X \Rightarrow X)^{z'}]}{yz' : B \Rightarrow (X \Rightarrow X)} \Rightarrow_{e} \quad \dfrac{[v : ((X \Rightarrow X) \Rightarrow B)^{v}] \quad [z' : (X \Rightarrow X)^{z'}]}{vz' : B} \Rightarrow_{e}}{yz'(vz') : X \Rightarrow X} \Rightarrow_{e}}{\lambda z'.yz'(vz') : (X \Rightarrow X) \Rightarrow (X \Rightarrow X)} \Rightarrow_{i}^{z'} \qquad \dfrac{[x : X^{x}]}{\lambda x.x : X \Rightarrow X} \Rightarrow_{i}^{x}}{(\lambda z'.yz'(vz'))\lambda x.x : X \Rightarrow X} \Rightarrow_{e}}{\lambda v.(\lambda z'.yz'(vz'))\lambda x.x : ((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)} \Rightarrow_{i}^{v}}{\lambda y.\lambda v.(\lambda z'.yz'(vz'))\lambda x.x : ((X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)) \Rightarrow ((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X))} \Rightarrow_{i}^{y} \qquad \dfrac{\dfrac{[z : (X \Rightarrow X)^{z}]}{\lambda b.z : B \Rightarrow X \Rightarrow X} \Rightarrow_{i}^{b}}{\lambda z.\lambda b.z : (X \Rightarrow X) \Rightarrow (B \Rightarrow X \Rightarrow X)} \Rightarrow_{i}^{z}$$
$$(\lambda y.\lambda v.(\lambda z'.yz'(vz'))\lambda x.x)\lambda z.\lambda b.z : ((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X) \Rightarrow_{e}$$

Thus, the λ-term decorating $\mathcal{D}$ is $t = (\lambda y.\lambda v.(\lambda z'.yz'(vz'))\lambda x.x)\lambda z.\lambda b.z$. All possible β-reduction steps from t are the following:



In any case, every β-reduction sequence eventually reaches the same β-normal term $\lambda v.\lambda x.x$. Note that $\lambda v.\lambda x.x$ is the decoration of the derivation $\mathcal{D}_0$ below, which is the derivation without redexes to which $\mathcal{D}$ eventually reduces

via cut-elimination steps (see Exercise 3 from day 1).

$$\dfrac{\dfrac{[x : X^x]}{\lambda x.x : X \Rightarrow X}\Rightarrow_i^x}{\lambda v.\lambda x.x : ((X \Rightarrow X) \Rightarrow B) \Rightarrow (X \Rightarrow X)}\Rightarrow_i^v$$

Exercise 3

Prove rigorously the following facts ($f^n x = \overbrace{f(\dots(f}^{n \text{ times } f}\, x)\dots)$ for any $n \in \mathbb{N}$):

1. $\lambda x.xx$ is untypable in Curry-style, $\lambda x^A.xx$ is untypable in Church-style for any type A;
2. in Church-style, $\lambda f^Y.\lambda x^X.f^n x$ is not typable for any $n > 0$ but $\lambda f^Y.\lambda x^X.x$ is typable;
3. $\lambda f.\lambda x.f^n x$ is typable in Curry-style, for all $n \in \mathbb{N}$.

Solution to Exercise 3

1. *Curry-style:* Suppose by absurd that $\lambda x.xx$ is typable in the simply typed λ-calculus in Curry-style. Then there would be a derivation $\mathcal{D}$ of $\lambda x.xx$. Its last rule is necessarily λ (because the term in the derivation is an abstraction), and its second to last rule is necessarily @ (because the body of the abstraction in the derivation is an application), and its leaves are necessarily var rules (because the proper subterms of the application are variables), hence $\mathcal{D}$ has the form below, for some types A, B, C.

$$\dfrac{\dfrac{\dfrac{}{x : A \vdash x : C \Rightarrow B}\,\mathsf{var} \qquad \dfrac{}{x : A \vdash x : C}\,\mathsf{var}}{x : A \vdash xx : B}\,@}{\vdash \lambda x.xx : A \Rightarrow B}\,\lambda$$

   To make $\mathcal{D}$ a valid derivation, the two instances of the rule var must be correct, thus $A = C \Rightarrow B$ and $A = C$ must hold, which implies that $C = C \Rightarrow B$, but this is impossible for any type B, C.

   Church-style: Suppose by absurd that $\lambda x^A.xx$ is typable in the simply typed λ-calculus in Church-style. Then there would be a derivation $\mathcal{D}$ of $\lambda x^A.xx$. Its last rule is necessarily λ abstracting a variable of type A (because the term in the derivation is an abstraction of type A), and its second to last rule is necessarily @ (because the body of the abstraction is an application), and its leaves are necessarily var rules (because the proper subterms of the application are variables), hence $\mathcal{D}$ has the form below, for some types B, C.

$$\dfrac{\dfrac{\dfrac{}{x : A \vdash x : C \Rightarrow B}\,\mathsf{var} \qquad \dfrac{}{x : A \vdash x : C}\,\mathsf{var}}{x : A \vdash xx : B}\,@}{\vdash \lambda x^A.xx : A \Rightarrow B}\,\lambda$$

   To make $\mathcal{D}$ a valid derivation, the two instances of the rule var must be correct, thus $A = C \Rightarrow B$ and $A = C$ must hold, which implies that $C = C \Rightarrow B$, but this is impossible for any type B, C.

2. The term $\lambda f^Y.\lambda x^X.x$ is typable in Church-style, as shown by the derivation below.

$$\dfrac{\dfrac{\dfrac{}{f : Y,\, x : X \vdash x : X}\,\mathsf{var}}{f : Y \vdash \lambda x^X.x : X \Rightarrow X}\,\lambda}{\vdash \lambda f^Y.\lambda x^X.x : Y \Rightarrow X \Rightarrow X}\,\lambda$$

   We prove by contradiction that $\lambda f^Y.\lambda x^X.f^n x$ is not typable in Church-style for any $n \in \mathbb{N}^+$. Since $n \in \mathbb{N}^+ = \mathbb{N} \setminus \{0\}$, then $f^n x = f(f^{n-1}x)$ where $n - 1 \in \mathbb{N}$. Suppose by absurd that $\lambda f^Y.\lambda x^X.f^n x$ is typable in the simply typed λ-calculus in Church-style. Then there would be a derivation $\mathcal{D}$ of $\lambda f^Y.\lambda x^X.f^n x$. Its two last rules are necessarily λ (because the term in the derivation is a double abstraction), and its third to last rule is necessarily @ (because the body of the double abstraction is the application $f(f^{n-1}x)$), and the left premise of the @ rule is necessarily a var rule (because the left subterm of the application is a variable), hence $\mathcal{D}$ has the form below, for some types A, B.

$$\dfrac{\dfrac{\dfrac{}{f : Y,\, x : X \vdash f : B \Rightarrow A}\,\mathsf{var} \qquad \begin{array}{c}\vdots\\ f : Y,\, x : X \vdash f^{n-1}x : B\end{array}}{f : Y,\, x : X \vdash f^n x : A}\,@}{f : Y \vdash \lambda x^X.x : X \Rightarrow A}\,\lambda}{\vdash \lambda f^Y.\lambda x^X.x : Y \Rightarrow X \Rightarrow A}\,\lambda$$

To make $\mathcal{D}$ a valid derivation, the left instance of the rule var must be correct, thus $Y = B \Rightarrow A$ must hold for some types A, B, but this is impossible because Y is a ground type.

3. We first prove the following.

   Fact. *For all $n \in \mathbb{N}$, there is a derivation of $f : X \Rightarrow X,\, x : X \vdash f^n x : X$ (in Curry-style and Church-style).*

   Proof. By induction on $n \in \mathbb{N}$. Cases:

   (a) $n = 0$: then, $f^0 x = x$ and hence the derivation below concludes.

   $$\dfrac{}{f : X \Rightarrow X,\, x : X \vdash x : X}\,\mathsf{var}$$

   (b) $n > 0$: then $f^n x = f(f^{n-1}x)$ and by induction hypothesis there is a derivation $\mathcal{D}$ of $f : X \Rightarrow X,\, x : X \vdash f^{n-1}x : X$. The derivation below concludes.

   $$\dfrac{\dfrac{}{f : X \Rightarrow X,\, x : X \vdash f : X \Rightarrow X}\,\mathsf{var} \qquad \begin{array}{c}\mathcal{D}\\ \vdots\\ f : X \Rightarrow X,\, x : X \vdash f^{n-1}x : X\end{array}}{f : X \Rightarrow X,\, x : X \vdash f^n x : X}\,@$$

   □

   We can now show that, for all $n \in \mathbb{N}$, the term $\lambda f.\lambda x.f^n x$ is typable in Curry-style. Indeed, by the fact above, there is a derivation $\mathcal{D}$ of $f : X \Rightarrow X,\, x : X \vdash f^n x : X$ for all $n \in \mathbb{N}$. The derivation below concludes:

   $$\dfrac{\dfrac{\dfrac{\begin{array}{c}\mathcal{D}\\ \vdots\\ f : X \Rightarrow X,\, x : X \vdash f^n x : X\end{array}}{f : X \Rightarrow X \vdash \lambda x.f^n x : X \Rightarrow X}\,\lambda}{\vdash \lambda f.\lambda x.f^n x : (X \Rightarrow X) \Rightarrow X \Rightarrow X}\,\lambda$$

Exercise 4

Prove that if t is typable in Church or Curry style, then so is every subterm of t.

Solution to Exercise 4

Let us recall that the *subterms* of a λ-term t are the elements of the set $\mathsf{subt}(t)$ defined below by induction on t:

$$\mathsf{subt}(x) = \{x\} \qquad \mathsf{subt}(\lambda x.t) = \{\lambda x.t\} \cup \mathsf{subt}(t) \qquad \mathsf{subt}(st) = \{st\} \cup \mathsf{subt}(s) \cup \mathsf{subt}(t)$$

We now prove that if a term t is typable in STLC in Curry style, then so is every subterm of t. The proof for t typable in STLC in Church style is analogous. We proceed by induction on t. Let $s \in \mathsf{subt}(t)$. Cases:

- If $s = t$ then s is typable by hypothesis.
- If $s \neq t = \lambda x.t'$ then $s \in \mathsf{subt}(t')$. Since t is typable in STLC in Curry style, then there is a derivation $\mathcal{D} \;\triangleright_{\mathrm{Curry}}\; \Gamma \vdash \lambda x.t' : A$ for some environment Γ and type A. Necessarily, $\mathcal{D}$ has the form below, with $A = B \Rightarrow C$ for some types B, C:

  $$\mathcal{D} \;=\; \dfrac{\begin{array}{c}\mathcal{D}'\\ \vdots\\ \Gamma, x : B \vdash t' : C\end{array}}{\Gamma \vdash \lambda x.t' : B \Rightarrow C}\,\Rightarrow_i$$

  Hence, t' is typable. By induction hypothesis applied to t', s is typable.

- If $s \neq t = t_1 t_2$ then $s \in \mathsf{subt}(t_1) \cup \mathsf{subt}(t_2)$. Since t is typable in STLC in Curry style, then there is a derivation $\mathcal{D} \rhd_{\text{Curry}} \Gamma \vdash t_1 t_2 : A$ for some environment Γ and type A. Necessarily, $\mathcal{D}$ has the form below:

$$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma \vdash t_1 : B \Rightarrow A & \Gamma \vdash t_2 : B\end{array}}{\Gamma \vdash t_1 t_2 : A}\Rightarrow_e$$

  Hence, t_1 and t_2 are typable. By induction hypothesis applied to t_1 or t_2 (depending on whether $s \in \mathsf{subt}(t_1)$ or $s \in \mathsf{subt}(t_2)$), s is typable.

This complete the proof.

Remark. *Note that the case $t = s$ includes the case where t is a variable, since the only subterm of a variable is the variable itself.*

Exercise 5

Prove rigorously the propositions on pp. 21 and 25, the lemma and theorems on p. 26 of the Lecture 2 slides.

Solution to Exercise 5

We first define some notions that will be used in the proofs. The *size* of a derivation in STLC is its number of typing rule instances. A derivation $\mathcal{D} \rhd_{\mathsf{STLC}} \Gamma \vdash t : A$ is *minimal* if there is no derivation $\mathcal{D}' \rhd_{\mathsf{STLC}} \Gamma \vdash t : A$ whose size is less than the size of $\mathcal{D}$.

Proposition.

1. (Weakening) *If $\Gamma \vdash_{\mathsf{STLC}} t : A$, then $\Gamma, x : B \vdash_{\mathsf{STLC}} t : A$, for any type B and variable $x \notin \mathsf{dom}(\Gamma)$.*
2. (Free variables) *If $\Gamma \vdash_{\mathsf{STLC}} t : A$, then $\mathsf{fv}(t) \subseteq \mathsf{dom}(\Gamma)$ and $\Gamma\restriction_{\mathsf{fv}(t)} \vdash_{\mathsf{STLC}} t : A$.*

Proof. We prove both statement for STLC in Curry style. The proof for Church style is analogous.

1. Let B be a type and $x \notin \mathsf{dom}(\Gamma)$. We prove the statement by induction on the size of a minimal derivation $\mathcal{D} \rhd_{\text{Curry}} \Gamma \vdash t : A$. Let us consider the last rule r of $\mathcal{D}$. Cases:

   - If $\mathsf{r} = \mathsf{var}$ then $\mathcal{D} = \dfrac{}{\Gamma', y : A \vdash y : A}\,\mathsf{var}$ with $\Gamma = \Gamma'$, $y : A$ and $t = y$. Since $x \notin \mathsf{dom}(\Gamma)$, the derivation $\mathcal{D}' = \dfrac{}{\Gamma', x : B, y : A \vdash y : A}\,\mathsf{var}$ is such that $\mathcal{D}' \rhd_{\text{Curry}} \Gamma, x : B \vdash t : A$.
   - If $\mathsf{r} = \lambda$ then

$$\mathcal{D} = \dfrac{\vdots\,\mathcal{D}_0}{\Gamma, y : C \vdash s : D}\,\lambda$$

     with $A = C \Rightarrow D$ and $t = \lambda y.s$; we can suppose without loss of generality that $y \notin \mathsf{dom}(\Gamma) \cup \{x\}$. By induction hypothesis applied to $\mathcal{D}_0$, there is $\mathcal{D}'_0 \rhd_{\text{Curry}} \Gamma, y : C, x : B \vdash s : D$. Hence, the derivation $\mathcal{D}' = \dfrac{\vdots\,\mathcal{D}'_0}{\Gamma, y : C, x : B \vdash s : D}\,\lambda$ is such that $\mathcal{D}' \rhd_{\text{Curry}} \Gamma, x : B \vdash t : A$.
   - If $\mathsf{r} = @$ then

$$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma \vdash t_1 : C \Rightarrow A & \Gamma \vdash t_2 : C\end{array}}{\Gamma \vdash t_1 t_2 : A}@$$

     with $t = t_1 t_2$. By induction hypothesis applied to $\mathcal{D}_1$ and $\mathcal{D}_2$, there are $\mathcal{D}'_1 \rhd_{\text{Curry}} \Gamma, x : B \vdash t_1 : C \Rightarrow A$ and $\mathcal{D}'_2 \rhd_{\text{Curry}} \Gamma, x : B \vdash t_2 : C$. Hence, the derivation $\mathcal{D}' = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}'_1 & \vdots\,\mathcal{D}'_2\\ \Gamma, x : B \vdash t_1 : C \Rightarrow A & \Gamma, x : B \vdash t_2 : C\end{array}}{\Gamma, x : B \vdash t_1 t_2 : A}@$ is such that $\mathcal{D}' \rhd_{\text{Curry}} \Gamma, x : B \vdash t : A$.

2. Proof by induction on the size of a minimal derivation $\mathcal{D} \triangleright_{\text{Curry}} \Gamma \vdash t : A$. Let us consider the last rule r of $\mathcal{D}$. Cases:

   - If $\mathsf{r} = \mathsf{var}$ then $\mathcal{D} = \dfrac{}{\Gamma', y : A \vdash y : A}\,\mathsf{var}$ with $\Gamma = \Gamma'$, $y : A$ and $t = y$. Hence, $\mathsf{fv}(t) = \{y\} \subseteq \mathsf{dom}(\Gamma)$ and $\Gamma \restriction_{\mathsf{fv}(t)} = y : A$, so the derivation $\mathcal{D}' = \dfrac{}{y : A \vdash y : A}\,\mathsf{var}$ is such that $\mathcal{D}' \triangleright_{\text{Curry}} \Gamma \restriction_{\mathsf{fv}(t)} \vdash t : A$.
   - If $\mathsf{r} = \lambda$ then
   $$\mathcal{D} = \dfrac{\begin{array}{c}\vdots\,\mathcal{D}_0\\ \Gamma, y : C \vdash s : D\end{array}}{\Gamma \vdash \lambda y.s : C \Rightarrow D}\,\lambda$$
   with $A = C \Rightarrow D$ and $t = \lambda y.s$; we can suppose without loss of generality that $y \notin \mathsf{dom}(\Gamma)$. By induction hypothesis applied to $\mathcal{D}_0$, $\mathsf{fv}(s) \subseteq \mathsf{dom}(\Gamma) \cup \{y\}$ and there is $\mathcal{D}'_0 \triangleright_{\text{Curry}} (\Gamma, y : C) \restriction_{\mathsf{fv}(s)} \vdash s : D$. Hence, $\mathsf{fv}(t) = \mathsf{fv}(s) \setminus \{y\} \subseteq \mathsf{dom}(\Gamma)$, and the derivation
   $$\mathcal{D}' = \dfrac{\begin{array}{c}\vdots\,\mathcal{D}'_0\\ (\Gamma, y : C) \restriction_{\mathsf{fv}(s)} \vdash s : D\end{array}}{\Gamma \restriction_{\mathsf{fv}(s)} \vdash \lambda y.s : C \Rightarrow D}\,\lambda$$
   is such that $\mathcal{D}' \triangleright_{\text{Curry}} \Gamma \restriction_{\mathsf{fv}(t)} \vdash t : A$, since $y \notin \mathsf{dom}(\Gamma) \supseteq \mathsf{fv}(t)$.
   - If $\mathsf{r} = @$ then
   $$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma \vdash t_1 : C \Rightarrow A & \Gamma \vdash t_2 : C\end{array}}{\Gamma \vdash t_1 t_2 : A}\,@$$
   with $t = t_1 t_2$. By induction hypothesis applied to $\mathcal{D}_1$ and $\mathcal{D}_2$, $\mathsf{fv}(t_1) \subseteq \mathsf{dom}(\Gamma)$ and $\mathsf{fv}(t_2) \subseteq \mathsf{dom}(\Gamma)$ and there are $\mathcal{D}'_1 \triangleright_{\text{Curry}} \Gamma \restriction_{\mathsf{fv}(t_1)} \vdash t_1 : C \Rightarrow A$ and $\mathcal{D}'_2 \triangleright_{\text{Curry}} \Gamma \restriction_{\mathsf{fv}(t_2)} \vdash t_2 : C$. Hence, $\mathsf{fv}(t) = \mathsf{fv}(t_1) \cup \mathsf{fv}(t_2) \subseteq \mathsf{dom}(\Gamma)$. By Item 1, there are $\mathcal{D}''_1 \triangleright_{\text{Curry}} \Gamma \restriction_{\mathsf{fv}(t)} \vdash t_1 : C \Rightarrow A$ and $\mathcal{D}''_2 \triangleright_{\text{Curry}} \Gamma \restriction_{\mathsf{fv}(t)} \vdash t_2 : C$. Therefore, the derivation $\mathcal{D}' = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}''_1 & \vdots\,\mathcal{D}''_2\\ \Gamma \restriction_{\mathsf{fv}(t)} \vdash t_1 : C \Rightarrow A & \Gamma \restriction_{\mathsf{fv}(t)} \vdash t_2 : C\end{array}}{\Gamma \restriction_{\mathsf{fv}(t)} \vdash t_1 t_2 : A}\,@$ is such that $\mathcal{D}' \triangleright_{\text{Curry}} \Gamma \restriction_{\mathsf{fv}(t)} \vdash t : A$. □

Proposition. *1.* (From Church to Curry) *If* $\Gamma \vdash_{\text{Church}} t : A$, *then* $\Gamma \vdash_{\text{Curry}} \lceil t \rceil : A$.

2. (From Curry to Church) *If* $\Gamma \vdash_{\text{Curry}} t : A$ *then* $\Gamma \vdash_{\text{Church}} t' : A$ *for some* t' *in Church-style such that* $\lceil t' \rceil = t$.

Proof. 1. Proof by induction on the size of a minimal derivation $\mathcal{D} \triangleright_{\text{Church}} \Gamma \vdash t : A$. Let us consider the last rule r of $\mathcal{D}$. Cases:

   - If $\mathsf{r} = \mathsf{var}$ then $\mathcal{D} = \dfrac{}{\Gamma', y : A \vdash y : A}\,\mathsf{var}$ with $\Gamma = \Gamma'$, $y : A$ and $t = y = \lceil t \rceil$. Hence $\mathcal{D} \triangleright_{\text{Curry}} \Gamma \vdash \lceil t \rceil : A$.
   - If $\mathsf{r} = \lambda$ then
   $$\mathcal{D} = \dfrac{\begin{array}{c}\vdots\,\mathcal{D}_0\\ \Gamma, y : C \vdash s : D\end{array}}{\Gamma \vdash \lambda y^C.s : C \Rightarrow D}\,\lambda$$
   with $A = C \Rightarrow D$ and $t = \lambda y^C.s$; we can suppose without loss of generality that $y \notin \mathsf{dom}(\Gamma)$. By induction hypothesis applied to $\mathcal{D}_0$, there is $\mathcal{D}'_0 \triangleright_{\text{Curry}} \Gamma, y : C \vdash \lceil s \rceil : D$. Hence, the derivation
   $$\mathcal{D}' = \dfrac{\begin{array}{c}\vdots\,\mathcal{D}'_0\\ \Gamma, y : C \vdash \lceil s \rceil : D\end{array}}{\Gamma \vdash \lambda y.\lceil s \rceil : C \Rightarrow D}\,\lambda$$
   is such that $\mathcal{D}' \triangleright_{\text{Curry}} \Gamma \vdash \lceil t \rceil : A$, since $\lceil t \rceil = \lambda y.\lceil s \rceil$.

- If $\mathsf{r} = @$ then

$$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma \vdash t_1 : C \Rightarrow A & \Gamma \vdash t_2 : C\end{array}}{\Gamma \vdash t_1 t_2 : A}\,@$$

with $t = t_1 t_2$. By induction hypothesis applied to $\mathcal{D}_1$ and $\mathcal{D}_2$, there are $\mathcal{D}'_1 \triangleright_{\text{Curry}} \Gamma \vdash \lceil t_1 \rceil : C \Rightarrow A$ and $\mathcal{D}'_2 \triangleright_{\text{Curry}} \Gamma \vdash \lceil t_2 \rceil : C$. Therefore, the derivation $\mathcal{D}' = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}'_1 & \vdots\,\mathcal{D}'_2\\ \Gamma \vdash t_1 : C \Rightarrow A & \Gamma \vdash t_2 : C\end{array}}{\Gamma \vdash \lceil t_1 \rceil \lceil t_2 \rceil : A}\,@$ is such that $\mathcal{D}' \triangleright_{\text{Curry}} \Gamma \vdash \lceil t \rceil : A$, since $\lceil t \rceil = \lceil t_1 \rceil \lceil t_2 \rceil$.

2. Proof by induction on the size of a minimal derivation $\mathcal{D} \triangleright_{\text{Curry}} \Gamma \vdash t : A$. Let us consider the last rule r of $\mathcal{D}$. Cases:

   - If $\mathsf{r} = \mathsf{var}$ then $\mathcal{D} = \dfrac{}{\Gamma', y : A \vdash y : A}\,\mathsf{var}$ with $\Gamma = \Gamma'$, $y : A$ and $t = y$. Let $t' \coloneqq y$. Hence, $\lceil t' \rceil = y = t$ and $\mathcal{D} \triangleright_{\text{Church}} \Gamma \vdash t' : A$.
   - If $\mathsf{r} = \lambda$ then

$$\mathcal{D} = \dfrac{\vdots\,\mathcal{D}_0}{\Gamma, y : C \vdash s : B}{\Gamma \vdash \lambda y.s : C \Rightarrow B}\,\lambda$$

   with $A = C \Rightarrow B$ and $t = \lambda y.s$; we can suppose without loss of generality that $y \notin \mathsf{dom}(\Gamma)$. By induction hypothesis applied to $\mathcal{D}_0$, there is $\mathcal{D}'_0 \triangleright_{\text{Church}} \Gamma, y : C \vdash s' : B$ for some s' in Church style such that $\lceil s' \rceil = s$. Let $t' \coloneqq \lambda y^C.s'$. Hence, the derivation

$$\mathcal{D}' = \dfrac{\vdots\,\mathcal{D}'_0}{\Gamma, y : C \vdash s' : B}{\Gamma \vdash \lambda y^C.s' : C \Rightarrow B}\,\lambda$$

   is such that $\mathcal{D}' \triangleright_{\text{Church}} \Gamma \vdash t' : A$, and $\lceil t' \rceil = \lambda y.\lceil s' \rceil = \lambda y.s = t$.
   - If $\mathsf{r} = @$ then

$$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma \vdash t_1 : C \Rightarrow A & \Gamma \vdash t_2 : C\end{array}}{\Gamma \vdash t_1 t_2 : A}\,@$$

   with $t = t_1 t_2$. By induction hypothesis applied to $\mathcal{D}_1$ and $\mathcal{D}_2$, there are $\mathcal{D}'_1 \triangleright_{\text{Church}} \Gamma \vdash t'_1 : C \Rightarrow A$ and $\mathcal{D}'_2 \triangleright_{\text{Church}} \Gamma \vdash t'_2 : C$ for some t'_1 and t'_2 in Church style such that $\lceil t'_1 \rceil = t_1$ and $\lceil t'_2 \rceil = t_2$. Let $t' \coloneqq t'_1 t'_2$. Thus, the derivation $\mathcal{D}' = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}'_1 & \vdots\,\mathcal{D}'_2\\ \Gamma \vdash t'_1 : C \Rightarrow A & \Gamma \vdash t'_2 : C\end{array}}{\Gamma \vdash t'_1 t'_2 : A}\,@$ is such that $\mathcal{D}' \triangleright_{\text{Church}} \Gamma \vdash t' : A$, and $\lceil t' \rceil = \lceil t'_1 \rceil \lceil t'_2 \rceil = t_1 t_2 = t$. □

Proposition (Uniqueness)**.** *If $\mathcal{D} \triangleright_{\text{Church}} \Gamma \vdash t : A$ and $\mathcal{D}' \triangleright_{\text{Church}} \Gamma \vdash t : A'$, then $A = A'$ and $\mathcal{D} = \mathcal{D}'$.*

Proof. By induction on t. Cases:

- If $t = y$ then $\mathcal{D} = \dfrac{}{\Gamma_0, y : A \vdash y : A}\,\mathsf{var} = \mathcal{D}'$ with $\Gamma = \Gamma_0$, $y : A$ and $t = y$.
- If $t = \lambda x.t_0$ then

$$\mathcal{D} = \dfrac{\vdots\,\mathcal{D}_0}{\Gamma, y : C \vdash t_0 : B}{\Gamma \vdash \lambda y^C.t_0 : C \Rightarrow B}\,\lambda \qquad \mathcal{D}' = \dfrac{\vdots\,\mathcal{D}'_0}{\Gamma, y : C' \vdash t_0 : B'}{\Gamma \vdash \lambda y^{C'}.t_0 : C' \Rightarrow B'}\,\lambda$$

  with $A = C \Rightarrow B$ and $A' = C' \Rightarrow B'$; we can suppose without loss of generality that $y \notin \mathsf{dom}(\Gamma)$. As $\lambda y^C.t_0 = t = \lambda y^{C'}.t_0$, then $C = C'$. By induction hypothesis applied to t_0, $B = B'$ and $\mathcal{D}_0 = \mathcal{D}'_0$. Hence, $A = C \Rightarrow B = C' \Rightarrow B' = A'$ and so $\mathcal{D} = \mathcal{D}'$.

- If $t = t_1 t_2$ then

$$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma \vdash t_1 : C \Rightarrow A & \Gamma \vdash t_2 : C\end{array}}{\Gamma \vdash t_1 t_2 : A}\,@ \qquad \mathcal{D}' = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}'_1 & \vdots\,\mathcal{D}'_2\\ \Gamma \vdash t_1 : C' \Rightarrow A' & \Gamma \vdash t_2 : C'\end{array}}{\Gamma \vdash t_1 t_2 : A'}\,@$$

  By induction hypothesis applied to t_1 and t_2, $C \Rightarrow A = C' \Rightarrow A'$, $C = C'$, $\mathcal{D}_1 = \mathcal{D}'_1$ and $\mathcal{D}_2 = \mathcal{D}'_2$. Therefore, $A = A'$ and $\mathcal{D} = \mathcal{D}'$. □

Lemma (Substitution)**.** *If* $\Gamma, x : B \vdash_{\mathsf{STLC}} t : A$ *and* $\Gamma \vdash_{\mathsf{STLC}} s : B$*, then* $\Gamma \vdash_{\mathsf{STLC}} t\{s/x\} : A$.

Proof. We prove both statement for STLC in Curry style. The proof for Church style is analogous. Proof by structural induction t. Cases:

- If $t = x$ then $t\{s/x\} = s$. As $\Gamma, x : B \vdash_{\text{Curry}} t : A$, then $x \notin \mathsf{dom}(\Gamma)$ and $A = B$. Hence, the hypothesis $\Gamma \vdash_{\text{Curry}} s : B$ means that $\Gamma \vdash_{\text{Curry}} t\{s/x\} : A$.
- If $t = y \neq x$ then $t\{s/x\} = y = t$. As $\Gamma, x : B \vdash_{\text{Curry}} t : A$, then $\Gamma(y) = A$ and so $\Gamma \vdash_{\text{Curry}} t\{s/x\} : A$ by applying the var rule.
- If $t = \lambda y.t_0$ then we can suppose without loss of generality that $y \notin \mathsf{fv}(s) \cup \{x\}$, thus $t\{s/x\} = \lambda y.(t_0\{s/x\})$. As $\Gamma, x : B \vdash_{\text{Curry}} t : A$, then $A = C \Rightarrow D$ and $\Gamma, x : B, y : C \vdash_{\text{Curry}} t_0 : D$. By induction hypothesis applied to t_0, there is a derivation $\mathcal{D}_0 \triangleright_{\text{Curry}} \Gamma, y : C \vdash_{\text{Curry}} t_0\{s/x\} : D$. Therefore, the derivation

$$\mathcal{D} = \dfrac{\begin{array}{c}\vdots\,\mathcal{D}_0\\ \Gamma, y : C \vdash t_0\{s/x\} : D\end{array}}{\Gamma \vdash \lambda y.(t_0\{s/x\}) : C \Rightarrow D}\,\lambda$$

  is such that $\mathcal{D} \triangleright_{\text{Curry}} \Gamma \vdash t\{s/x\} : A$.
- If $t = t_1 t_2$ then $t\{s/x\} = t_1\{s/x\}t_2\{s/x\}$. As $\Gamma, x : B \vdash_{\text{Curry}} t : A$, then $\Gamma, x : B \vdash_{\text{Curry}} t_1 : C \Rightarrow A$ and $\Gamma, x : B \vdash_{\text{Curry}} t_2 : C$. By induction hypothesis applied to t_1 and t_2, there are derivations $\mathcal{D}_1 \triangleright_{\text{Curry}} \Gamma \vdash_{\text{Curry}} t_1\{s/x\} : C \Rightarrow A$ and $\mathcal{D}_2 \triangleright_{\text{Curry}} \Gamma \vdash_{\text{Curry}} t_2\{s/x\} : C$. Therefore, the derivation

$$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots\,\mathcal{D}_2\\ \Gamma \vdash t_1\{s/x\} : C \Rightarrow A & \Gamma \vdash t_2\{s/x\} : C\end{array}}{\Gamma \vdash t_1\{s/x\}t_2\{s/x\} : A}\,@$$

  is such that $\mathcal{D} \triangleright_{\text{Curry}} \Gamma \vdash t\{s/x\} : A$. □

Theorem (Subject reduction)**.** *If* $\Gamma \vdash_{\mathsf{STLC}} t : A$ *and* $t \to_\beta s$*, then* $\Gamma \vdash_{\mathsf{STLC}} s : A$.

Proof. We prove both statement for STLC in Curry style. The proof for Church style is analogous. Proof by structural induction on t. Cases:

- If $t = x$ then there is no s such that $t \to_\beta s$, and the statement is vacuously true.
- If $t = \lambda x.t_0$, then $A = B \Rightarrow C$ and we can suppose without loss of generality that $x \notin \mathsf{dom}(\Gamma)$. Thus, $t \to_\beta \lambda x.s_0 = s$ with $t_0 \to_\beta s_0$, and $\Gamma, x : B \vdash_{\text{Curry}} t_0 : C$. By induction hypothesis applied to t_0, there is a derivation $\mathcal{D}_0 \triangleright_{\text{Curry}} \Gamma, x : B \vdash s_0 : C$. Therefore, the derivation

$$\mathcal{D} = \dfrac{\begin{array}{c}\vdots\,\mathcal{D}_0\\ \Gamma, x : B \vdash s_0 : C\end{array}}{\Gamma \vdash \lambda x.s_0 : B \Rightarrow C}\,\lambda$$

  is such that $\mathcal{D} \triangleright_{\text{Curry}} \Gamma \vdash s : A$.
- If $t = t_1 t_2$, then $\Gamma \vdash_{\text{Curry}} t_1 : B \Rightarrow A$ and $\Gamma \vdash_{\text{Curry}} t_2 : B$. There are three subcases:
  - If $t_1 = \lambda x.t'_1$ (we can suppose without loss of generality that $x \notin \mathsf{dom}(\Gamma)$) and $t \to_\beta t'_1\{t_2/x\} = s$, then $\Gamma, x : B \vdash_{\text{Curry}} t'_1 : A$. By the substitution lemma above, $\Gamma \vdash_{\text{Curry}} t'_1\{t_2/x\} : A$ and so $\Gamma \vdash_{\text{Curry}} s : A$.

– If $t \to_\beta t_1' t_2 = s$ with $t_1 \to_\beta t_1'$, then by induction hypothesis applied to t_1, there is a derivation $\mathcal{D}_1 \triangleright_{\text{Curry}} \Gamma \vdash t_1' : B \Rightarrow A$. Therefore, the derivation

$$\mathcal{D} = \dfrac{\begin{array}{cc}\vdots\,\mathcal{D}_1 & \vdots \\ \Gamma \vdash t_1' : C & \Gamma \vdash t_2 : C\end{array}}{\Gamma \vdash t_1' t_2 : A}\,@$$

is such that $\mathcal{D} \triangleright_{\text{Curry}} \Gamma \vdash s : A$.

– If $t \to_\beta t_1 t_2' = s$ with $t_2 \to_\beta t_2'$, then the proof is analogous to the previous subcase. □